

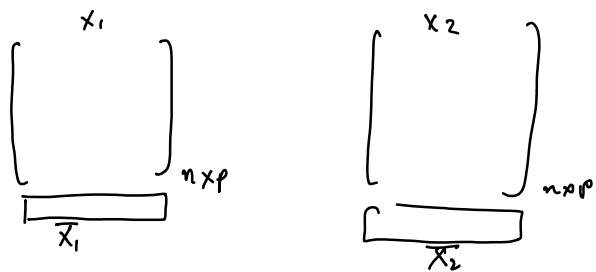
7 oct

07 October 2025 10:09

Paired t-test

Same subject on which experiment is conducted with p -outcomes $[N_p(\mu, \Sigma)]$
If I conduct this experiment 2 times under same setup.

both unknown



$$D_j = X_{1(j)} - X_{2(j)} \quad 1 \leq j \leq n$$

$$p \times 1 \quad p \times 1 \quad p \times 1$$

$$\bar{D} = \bar{X}_1 - \bar{X}_2 \quad (p \times 1) = \frac{1}{n} \sum_{j=1}^n D_j$$

$$H_0: \mu_1 - \mu_2 = 0 \text{ (or any number)}$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

$$H_0: \mu_1 - \mu_2 = \delta$$

$$H_1: \mu_1 - \mu_2 \neq \delta$$

under H_0 :

$$T^2 = n \bar{D}^T S_D^{-1} \bar{D}$$

$$(1 \times p) \quad (p \times p) \quad (p \times 1)$$

under H_0 :

$$T^2 = n (\bar{D} - \delta)^T S_D^{-1} (\bar{D} - \delta)$$

T_1
 $\vdots$
 T_q

The q -tests may be in specific order

$$H_0: \mu_1 = \mu_2 = \dots = \mu_q$$

$$H_1: \text{at least one of them different}$$

$$\begin{matrix} q-1 \\ \text{eqns} \end{matrix} \left\{ \begin{matrix} \mu_1 - \mu_2 = 0 \\ \mu_2 - \mu_3 = 0 \\ \vdots \\ \mu_{q-1} - \mu_q = 0 \end{matrix} \right.$$

$$\begin{matrix} \sum \\ (q-1) \times q \end{matrix} \mu_i \rightarrow q \times 1 = 0$$

define $Y_j = C X_j$
random vec. $(q-1) \times 1$ $\uparrow$ random $(q \times 1)$
 $(q-1) \times q$

if $X \sim N_q(\mu, \Sigma)$
$$Y_j \sim N_{q-1}(C\mu, C\Sigma C^T)$$

we can now apply hotelling T^2 -test on Y

$$T^2 = n (\bar{Y} - 0)^T (C S C^T)^{-1} (\bar{Y} - 0)$$

$$= n (C \bar{X})^T (C S C^T)^{-1} (C \bar{X})$$

$$= n \bar{X}^T C^T (C^T)^{-1} S^{-1} C^{-1} C \bar{X}$$

$$= n \bar{X}^T S^{-1} \bar{X}$$

$$\mu = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_q \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \end{bmatrix}$$

$$(q-1) \times q$$

row sum = 0
such matrix are called contrast matrix
A contrast vector have sum total of its components = 0
a matrix is contrast matrix if each row vector is contrast vector

$$\frac{(n-1)(q-1)}{n-q+1} F_{q-1, n-q+1, \alpha}$$

Q $n=4$

3 questions asked at 3 different time points and response time is recorded

$v=3$

	$Q_1 (t_1 \text{ sec})$	$Q_1 (t_2)$	$Q_1 (t_3)$
1	10	8	5
2	12	9	7
3	11	6	4
4	9	7	3

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1 : at least one of them diff.

$$C = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\bar{x} = \begin{bmatrix} 10.5 \\ 7.5 \\ 4.75 \end{bmatrix}$$

$$C\bar{x} = \begin{bmatrix} 3 \\ 2.75 \end{bmatrix}$$

$$S = \begin{bmatrix} 1.667 & 1 & 1.5833 \\ & 1.667 & 1.25 \\ & & 2.25 \end{bmatrix}$$

$$CSC^T = \begin{bmatrix} 1.334 & -1 \\ -1 & 2.4167 \end{bmatrix}$$

$$\left\{ \begin{array}{l} t^2 = 86.98, \quad F = 19 \times \frac{(n-1)(v-1)}{n-v+1} \\ \sim 57 \end{array} \right.$$

Reject H_0